

ITA0614 – MACHINE LEARNING

ASSESSMENT CO2 AT1

1. A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is 0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 marks)

Given:

Input 1 (Study Hours) $x_1 = 2$

Input 2 (Attendance) $x_2 = 1$

Weight for Study Hours $w_1 = 0.6$

Weight for Attendance $w_2 = 0.4$

Bias, $b = -0.5$

Calculate the Net Input

The net input to the perceptron is calculated as:

$$\text{Net Input} = (w_1 \times x_1) + (w_2 \times x_2) + b$$

Substitute the given values:

$$= (0.6 \times 2) + (0.4 \times 1) + (-0.5)$$

$$= 1.2 + 0.4 - 0.5$$

$$= 1.1$$

$$\text{Net Input} = 1.1$$

Apply the Step Activation Function

The step activation function checks whether the net input is positive or negative.

- If **Net Input** ≥ 0 , then **Output** = **1**
- If **Net Input** < 0 , then **Output** = **0**

Net Input=1.1

Since,

$1.1 > 0$

Conclusion

- **Output = 1**
 - A perceptron output of **1** indicates that the student **passes**.
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2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)

Given:

Initial weight, $w = 0.5$

Learning rate, $\eta = 0.1$

Input, $x = 2$

Actual output, $y = 0.6$

Target output, $t = 1$

Calculate the Error

The error is calculated as:

$$\text{Error} = \text{Target Output} - \text{Actual Output}$$

$$= 1 - 0.6$$

$$= 0.4$$

$$\text{Error} = 0.4$$

Calculate the Weight Update (Delta Rule)

The delta rule is:

$$\Delta w = \eta \times \text{Error} \times x$$

Substitute the given values:

$$\Delta w = 0.1 \times 0.4 \times 2$$

$$= 0.08$$

$$\text{Weight Update } (\Delta w) = 0.08$$

Calculate the New Updated Weight

The new weight is:

$$\text{New Weight} = \text{Old Weight} + \Delta w$$

$$= 0.5 + 0.08$$

$$= 0.58$$

$$\text{New Updated Weight} = 0.58$$

Final Answer

$$\text{Error} = 0.4$$

$$\text{Weight Update } (\Delta w) = 0.08$$

$$\text{New Updated Weight} = 0.58$$